

Development and penta-metric evaluation of a virtual interview simulator

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ABSTRACT

The virtual reality interview training system (VRITS) can provide a manageable training approach for candidates who tend to be very nervous during interviews; yet, the major anxiety stimulating elements remain unknown. By developing the VRITS and analyzing people's anxiety levels with an orthogonal experiment, we investigated five factors. Results indicate that TYPE OF INTERVIEW QUESTIONS plays a major role in the interviewee's anxiety. Secondly, LEVEL OF REALISM and PREPARATION both have some degree of influence. Lastly, INTERROGATOR'S ATTITUDE and TIMED OR UNTIMED ANSWERS have little to no impact. This work contributes towards cues for designing future VRITS.

Index Terms: I.3.7 [Three-Dimensional Graphics and Realism]: Virtual reality; H.5.2 [User Interfaces]: Training, help, and documentation

1 INTRODUCTION

Researchers have conducted extensive research on systems for VR interview training and evaluation. For example, VRITS have been designed for high-anxiety populations like people with Autism Spectrum Disorder (ADS) and former convicts, for shy college students and autistic individuals to practice interviewing skills in a less anxiety-inducing virtual context [1]. Nevertheless, most studies today validate and evaluate an entire product without examining the multiple factors separately. As a result, not only do we know very little about the significant elements that trigger interview anxiety, but we also lack a solid understanding of how each factor affects the interviewee's performance. Therefore, a study on factors contributing to people's interview anxiety is required to create successful training programs and develop tailored therapy approaches.

Previous studies have found that visual display [5], interview questions [3], interviewer's attitude [4], timing [6] and preparation [2] can all have an impact on interviewee's anxiety. As such, we developed the VRITS where the above factors were introduced and investigated within an orthogonal design to examine their significance separately. First, we measured the interviewee's electrodermal activity (EDA) which has been associated with anxiety during the interview. Next, we collected the anxiety level of the interviewees through a self-assessment questionnaire. Then, with the mixed-effects model and

the associated post-hoc analysis, we found that various factors influenced interviewees' anxiety, among which TYPE OF INTERVIEW QUESTIONS had the most significant impact. The proposed work aims to identify anxiety-stimulating factors during job interviews in virtual reality and provides detailed insights into designing future VRITS.

2 VRITS FRAMEWORK

Our VRITS was developed in Unity3D, and the avatars were designed using *Ready Player Me*¹ and *Avatar SDK*² animated from *Mixamo*. It was then built as apk files and was run on the Meta Oculus Quest 2 headset. The virtual environment was created similar to the physical environment for providing a close to real experience, as presented in Fig. 1.



Figure 1: Real interview site (L) & virtual scenario built in Unity3D (R).

For the multi-factors design of VRITS, we identified five independent variables that potentially affect the interviewee's anxiety levels during a job interview in virtual reality: (I) LEVEL OF REALISM(4 levels), which corresponded to the levels of visual display of the interviewer from the least to the most immersive, including an interviewer presented on a laptop computer, a low-poly and cartoon-like 3D avatar in VR, a realistic 3D human avatar with high fidelity in VR and a face-to-face real human interviewer on site, Fig. 2; (II) TYPE OF INTERVIEW QUESTIONS(2 levels) corresponded to two categories of questions: professional inquiries and personnel interview questions; (III) INTERROGATOR'S ATTITUDE(2 levels) corresponded to the interviewer's attitude with one being positive and another negative; (IV) TIMED OR UNTIMED ANSWERS(2 levels) corresponded to whether to time each answer of the interviewee; Furthermore, (V) WITH OR WITHOUT PREPARATION(2 levels) corresponded to whether the interviewee is given 5 minutes to prepare for the interview before it begins. In order to determine the relative importance of these five variables, we constructed a mixed orthogonal table ($L_8(4^1 \times 2^4)$) with all these five variables in a total of eight sets of conditions (Table 1).

The flow chart of the experimental protocol is represented in Fig. 3. Participants were asked to fill out the Measure of Anxiety

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¹<https://readyplayer.me>

²<https://avatarsdk.com>

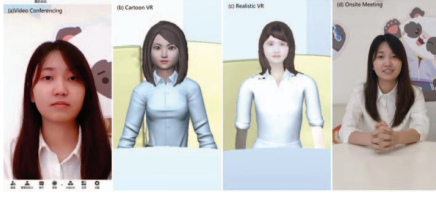


Figure 2: Least to most immersive(left to right):(a) video-conferencing, (b) cartoon VR, (c) realistic VR, (d) onsite meeting.

Table 1: Orthogonal design with multi-factors and mixed levels.

Experiment group	Interviewer attitude	Question type	Timing	Preparation	Realism
1	Positive	Personal	Yes	Yes	Video-conferencing
2	Positive	Personal	No	No	Realistic VR
3	Positive	Professional	Yes	No	Cartoon VR
4	Positive	Professional	No	Yes	Onsite meeting
5	Negative	Personal	Yes	No	Onsite meeting
6	Negative	Personal	No	Yes	Cartoon VR
7	Negative	Professional	Yes	Yes	Realistic VR
8	Negative	Professional	No	No	Video-conferencing

in Selection Interviews (MASI), and General Self-Efficacy (GSE) scales before being recruited to exclude extreme cases. In all eight conditions, participants were asked to wear an E4 wristband on their left wrist to record EDA data. At the end of each interview, the interviewees were asked to complete a self-assessment questionnaire to report their perceived anxiety.

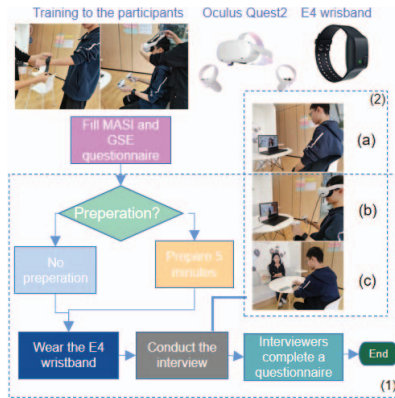


Figure 3: Flow chart of the experiment. (1) Repeat two times in one day (with a 12-hour interval in the morning and evening). Each participant performed the interviews in all eight conditions within four days. (2) Choose one of the experimental setups for each interview.

3 RESULTS

All participants completed the orthogonal experiment, so we opted for the mixed-effects model to analyze the data from repeated measures. We used the post-hoc analysis to compare the effect of different levels in one factor. Data analysis was performed in *R* and the significance level was set to .05.

Table 2 presents the results of mixed-effects model for analyzing the effect of five independent variables on interviewee's anxiety and ER-SCR³. *Question Type*, *Preparation* and *Realism* all have significant influence on *Anxiety* while *Interviewer attitude* and *Question type* both significantly affect *ER-SCR*. However, *Timekeeping* doesn't show any significant influence on interviewee's experience.

³It is the number of ER-SCR per seconds.

We further performed the post-hoc analysis for all factors, pairwise comparisons were performed using the t-test. A significant difference was found between negative and positive interviewer attitude regarding *ER-SCR* ($MD = .01, t_{133} = 2.10, p = .04, \eta_p^2 = .03$). Compared to *Personal question*, *Professional question* significantly leads to higher *Anxiety* ($MD = 24.07, t_{133} = 9.45, p < .01, \eta_p^2 = .40$) and less *ER-SCR* ($MD = -0.01, t_{133} = -2.07, p = .04, \eta_p^2 = .03$). *Preparation* is associated with less *Anxiety* ($MD = -5.95, t_{133} = -2.33, p = .02, \eta_p^2 = .04$) than that without *Preparation*. Compared with the *PC* interview, *Realistic VR* interview are linked to more *Anxiety* ($MD = -7.5, t_{133} = -2.08, p = .03, \eta_p^2 = .03$). Similarly, *Real person* interview leads to significantly lower *Anxiety* ($MD = -10.7, t_{133} = -2.97, p < .01, \eta_p^2 = .06$) than *Realistic VR*.

Table 2: Analysis of factors that affecting interviewee's experience using mixed-effects model(NumDF=1, DenDF=133)

	Interviewer attitude	Question type	Time keeping	Preparation	Realism
Anxiety	SS=55.7	SS=22032.2	SS=1.3	SS=1344.1	SS=2322.7
	F=23	F=89.31	F=.005	F=5.45	F=3.14
	p=.63	p<.01**	p=.94	p=.02*	p=.02*
	$\eta_p^2=.40$	$\eta_p^2=.00$	$\eta_p^2=.00$	$\eta_p^2=.04$	$\eta_p^2=.02$
ER-SCR (Gomboa2008)	SS=.0074	SS=.0072	SS=.0034	SS=.0005	SS=.0014
	F=4.38	F=4.30	F=2.05	F=.32	F=.28
	p=.04*	p=.04*	p=.15	p=.57	p=.84
	$\eta_p^2=.03$	$\eta_p^2=.03$	$\eta_p^2=.02$	$\eta_p^2=.00$	$\eta_p^2=.00$

4 DISCUSSION AND CONCLUSION

The developed VRITS system can used as an alternative interview tools during job-seeking as we found similar phenomena as in previous work [2–6]. For example, professional questions may increase the interviewee's anxiety while adequate preparation before the interview can alleviate it. On the other hand, the fidelity of virtual avatars and timekeeping will not affect the anxiety level, but a negative attitude from the interviewer will impose more stress on the user.

This work investigated the potential causes of anxiety in job interviews within VRITS. Our study remedies previous studies' limitation of focusing on only one type of measurement tool on anxiety, either physiological data or psychological questionnaires. Furthermore, using an orthogonal experimental design on multi-factors within our VRITS is a groundwork for designing VRITS with a more comprehensive framework in the future.

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